AXIAL PISTON PUMPS / PRODUCT INFORMATION

P6 Axial Piston Pump

Compare the compact pump's delivered flow across the intended pressure range. Use the published curves at matching speed and pressure; separate maximum flow and pressure values do not establish one simultaneous operating point.

The P6 is the mid-range workhorse of the WaterQuest axial piston pump family, delivering up to 12 L/min (3.17 US gpm) at a continuous 160 bar (2,321 psi). It steps up from the P3 with two displacement options of 4.6 and 6.0 cc/rev (0.281 and 0.366 in³/rev), and is available in unboosted 2-port and boosted 3-port configurations for maximum installation flexibility.



P6 stainless steel axial piston water hydraulic pump

At a glance

Parameter / value	Parameter / value
Max Displacement 6.0 cc/rev (0.366 in³/rev)	Min Displacement 4.6 cc/rev (0.281 in³/rev)
Max Pressure 160 bar (2,321 psi) continuous	Max Flow 12 L/min (3.17 US gpm)
Max Input Power 3.8 kW (5.10 hp)	Weight 2.2 kg (4.85 lb)

Selection basis

Match the selected configuration to the working water, pressure, flow, temperature and duty. Maximum ratings and separate curve points do not establish one simultaneous operating point. Confirm installation and supplied configuration with WaterQuest Solutions.

Technical data

Specifications and selection data

Parameter	4.6 cc/rev (0.281 in³/rev) Unboosted	4.6 cc/rev (0.281 in³/rev) Boosted	6.0 cc/rev (0.366 in³/rev) Unboosted	6.0 cc/rev (0.366 in³/rev) Boosted
Displacement	4.6 cc/rev (0.281 in³/rev)	4.6 cc/rev (0.281 in³/rev)	6.0 cc/rev (0.366 in³/rev)	6.0 cc/rev (0.366 in³/rev)
Max Continuous Pressure	160 bar (2,321 psi)			
Max Water Flow	~7.4 L/min (1.95 US gpm) at 1800 rpm	~8.0 L/min (2.11 US gpm) at 2000 rpm	~9.7 L/min (2.56 US gpm) at 1800 rpm	12 L/min (3.17 US gpm) at 2000 rpm
Max RPM	1800 rpm	2000 rpm	1800 rpm	2000 rpm
Min / Max Inlet Pressure	Positive at all times; max 2.5 bar (36.3 psi) / 2.5 bar (36.3 psi) min supply		Positive at all times; max 2.5 bar (36.3 psi) / 2.5 bar (36.3 psi) min supply	
Max Input Power	~2.4 kW (3.22 hp) at 1750 rpm, 160 bar (2,321 psi)		3.8 kW (5.1 hp)	
Weight	2.2 kg (4.85 lb)			
Max Fluid Temperature	50°C (consult WQS above 50°C (122 °F))			
Min Fluid Temperature	2°C (35.6 °F) (antifreeze option available)			
Housing Material	316 Stainless Steel			
Pressure (P) Port	3/8 in (9.52 mm) BSPP			
T Port (Unboosted)	1/2 in (12.70 mm) BSPP (tank return)		1/2 in (12.70 mm) BSPP (tank return)	
T Port (Boosted)	3/8 in (9.52 mm) BSPP (case drain to sump)		3/8 in (9.52 mm) BSPP (case drain to sump)	
Shaft Diameter	8.99 / 8.97 mm (0.354 / 0.353 in)			
Shaft Key	3 x 3 mm (0.118 x 0.118 in)			
Outer Flange Diameter	36.99 / 36.97 mm (1.46 / 1.46 in)			
Body Outer Diameter	76 mm (2.99 in)			
Body Inner Diameter	58 mm (2.28 in)			
Body Length	89 mm (3.5 in)			
Shaft Section Length	34 mm (16 mm (0.63 in) key engagement)			
Mounting	2 x 5.5 mm (0.217 in) holes, 23.9 mm (0.941 in) PCD spacing			
Shaft Rotation	PC = CW, PA = CCW		PC = CW, PA = CCW / PB = either direction	
Inlet Filtration	10 µm (394 µin) nominal, 25 µm (984 µin) absolute, beta-10 = 75			
Certifications	Confirm applicability and supplied documentation: ATEX II 2GD ck T6, ISO 9001 FM 87247, F4N			

Configuration and ordering data

Code	Motor	Pressure	Flow @ 1500rpm
P6-060-M04	0.37 kW (0.496 hp)	10 to 20 bar (145 to 290 psi)	~8.6 L/min (2.27 US gpm)
P6-060-M05	0.55 kW (0.738 hp)	40 bar (580 psi)	~8.5 L/min (2.25 US gpm)
P6-060-M07	1.1 kW (1.48 hp)	60 to 80 bar (870 to 1,160 psi)	~8.2 L/min (2.17 US gpm)
P6-060-M08	1.5 kW (2.01 hp)	100 bar (1,450 psi)	~7.5 L/min (1.98 US gpm)
P6-060-M09	2.2 kW (2.95 hp)	120 to 140 bar (1,740 to 2,031 psi)	~7.2 L/min (1.9 US gpm)
P6-060-M10	3.0 kW (4.02 hp)	160 bar (2,321 psi)	~6.8 L/min (1.8 US gpm)

Selection and integration

All performance figures on clean tap water at 15°C (59 °F). The exact operating fluid must be specified on all enquiries and orders. 316 SS Material Test Reports available on request.

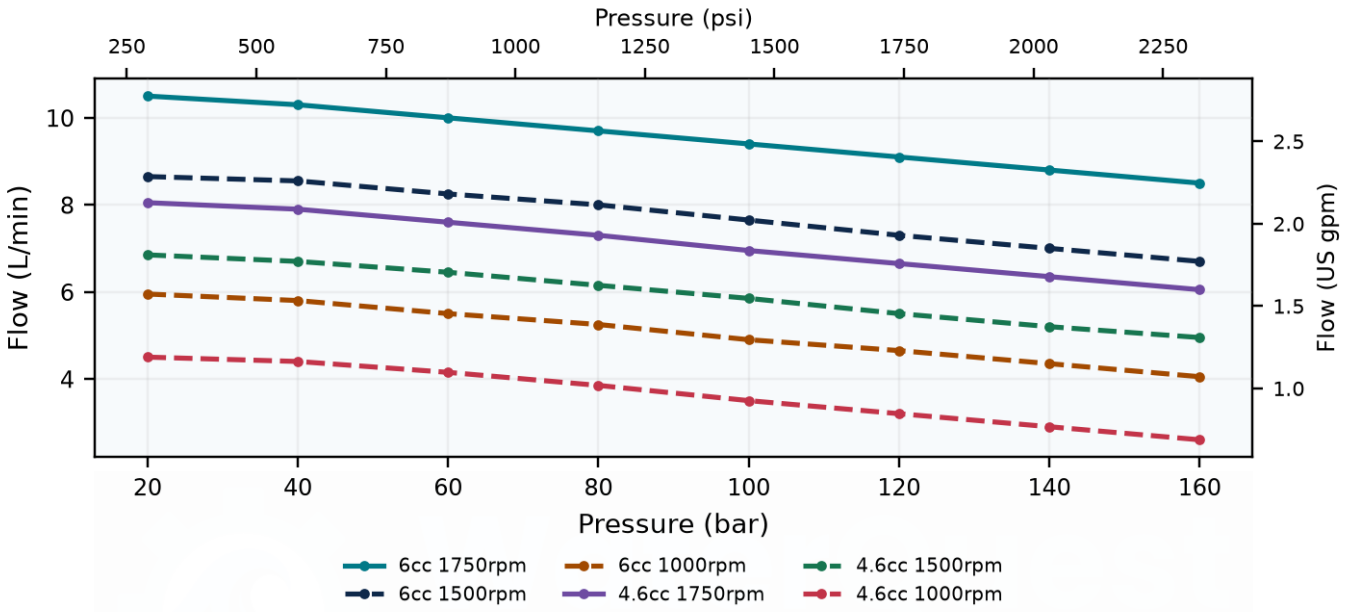
Selection and integration

Reference Engineering Calculators Reference Technical Glossary

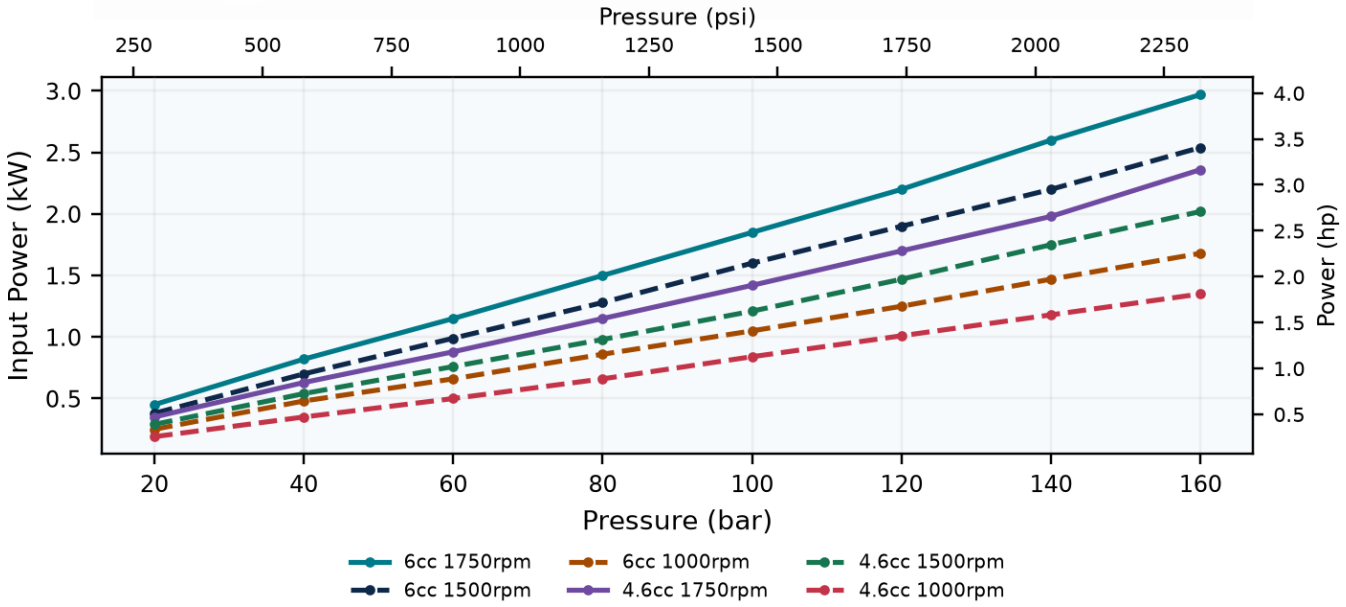
Performance curves

Reference curves reproduced from the product webpage. Curves retain the published data points and speed/displacement labels. Confirm guaranteed performance at the selected duty.

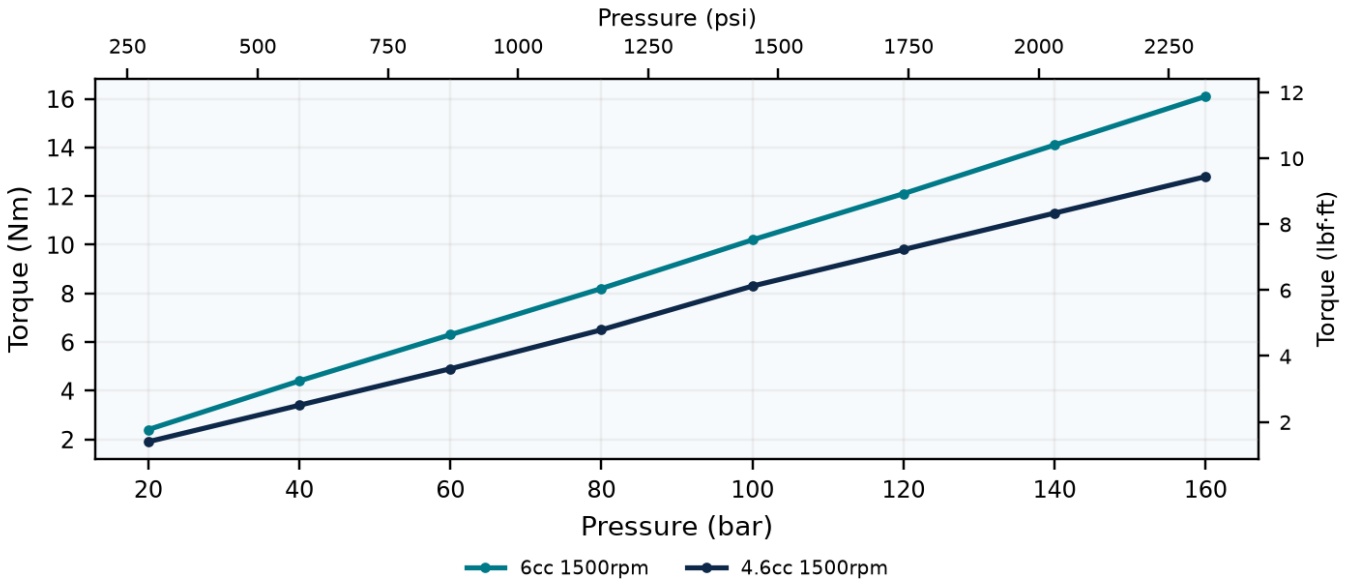
Flow (L/min) vs Pressure (bar)



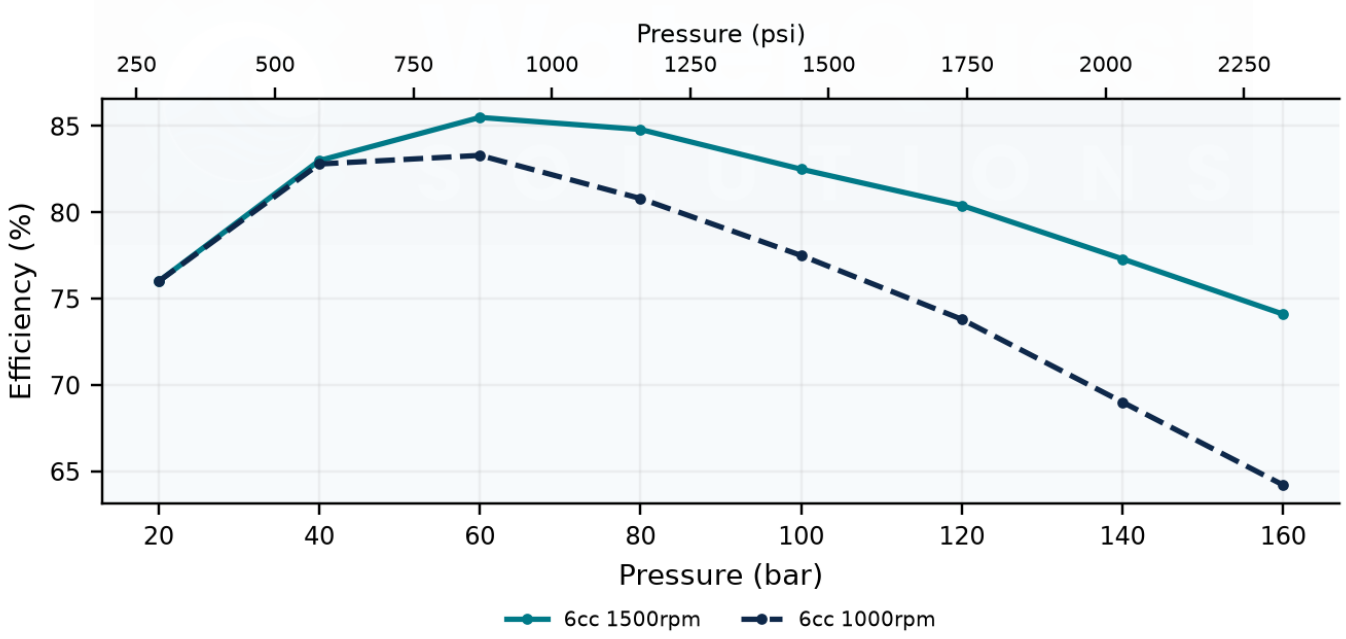
Input Power (kW) vs Pressure (bar)



Torque (Nm) vs Pressure (bar)



Efficiency (%) vs Pressure (bar)

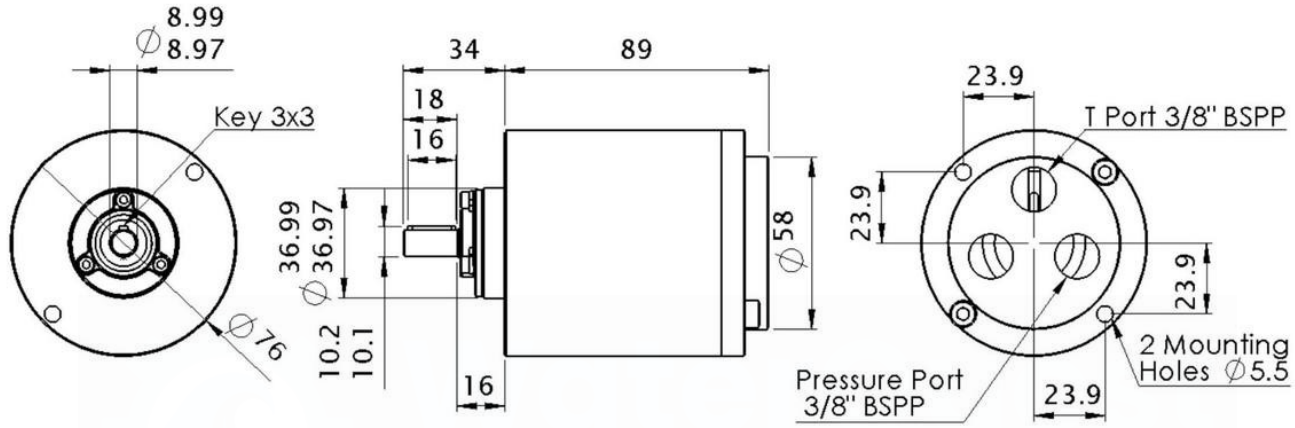


Drawings and reference visuals

Dimensions, ports and curves are reproduced from the webpage visuals. Use the original drawing units and confirm the current approved drawing before fabrication or installation.

PB160-6.0W and PB160-4.6W boosted P6 dimensional drawing, 3/8 inch (9.52 mm) BSPP pressure and case drain ports

PB160-6.0W & PB160-4.6W



PC160-6.0W and PA160-6.0W unboosted P6 dimensional drawing, 3/8 inch (9.52 mm) BSPP pressure port, 1/2 inch (12.70 mm) BSPP tank port

PC160-6.0W & PA160-6.0W

PC160-4.6W & PA160-4.6W

